

HEMA PRABA D

MCA Student | Full-Stack Developer (Aspiring) | IoT & Python Enthusiast

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PROFESSIONAL SUMMARY

Motivated MCA student with a strong academic record (CGPA 8.6) and hands-on experience in Python, JavaScript, IoT, and web development. Developed multiple end-to-end projects including an Expense Tracker Management System (HTML, CSS, JavaScript, MongoDB) and an IoT-based medical alerting system. Seeking a challenging software development role to apply technical expertise in building scalable, user-centric web applications.

TECHNICAL SKILLS

Programming Languages	Python, C, C++
Web Technologies	HTML5, CSS3, JavaScript (ES6+)
Mobile Development	Flutter (Dart)
Database	MongoDB, MySQL, Database Management
Embedded / IoT	ESP32, Arduino, HX711 Load Cell, SMTP Integration
Tools & Platforms	Git, VS Code, Arduino IDE
Core Competencies	Full-Stack Web Development, MongoDB, Data Visualization, Budget Management Systems, IoT Systems

EDUCATION

Master of Computer Applications (MCA) — Pursuing	2025 – 2027
Mepco Schlenk Engineering College, Sivakasi CGPA: 8.6	
Bachelor of Computer Applications (BCA)	2022 – 2025
The Standard Fireworks Rajaratnam College for Women CGPA: 8.73	
HSC — V.N.U.A.P.N.T MHSS, Sivakasi	2021 – 2022
Percentage: 87%	
SSLC — V.N.U.A.P.N.T MHSS, Sivakasi	2019 – 2020
Percentage: 80%	

PROJECTS

Expense Tracker Management System HTML5 CSS3 JavaScript MongoDB
- Built a full-stack personal finance web application with MongoDB backend, enabling users to manage income and expenses with real-time balance tracking. - Implemented Add Expense and Add Income modules with form validation, category tagging, and instant database persistence using MongoDB collections. - Developed View Expense and View Income dashboards with dynamic filtering, sorting, and paginated transaction history for streamlined financial review. - Integrated interactive Charts (bar, pie, and line) using JavaScript charting libraries to visualize spending patterns, income trends, and category-wise breakdowns. - Built a Budget Alert system that triggers real-time notifications when spending exceeds user-defined limits, promoting proactive financial management. - Designed a fully responsive UI using HTML5 and CSS3 with clean navigation across all six feature modules, ensuring a seamless cross-device user experience.

IoT-Based IV Bag Monitoring & Alerting System | *Python* | *ESP32/Arduino* | *SMTP* | *Hardware Sensors*

- Engineered a real-time IoT medical monitoring system using a load cell sensor to continuously track IV fluid levels in hospital settings.
- Automated multi-channel alerts (LED, buzzer, and SMTP email/video notifications) triggered when fluid level drops below critical threshold.
- Integrated HX711 ADC module for precise weight-to-volume conversion, ensuring accurate and reliable fluid level readings.
- Presented the research paper at ICAIC'25 international conference, demonstrating strong knowledge of embedded systems and healthcare IoT.

Library Management System | *HTML5* | *CSS3* | *Responsive Web Design*

- Designed and developed a responsive multi-page web application featuring user authentication (login), dashboard, and book issue/return workflows.
- Applied mobile-first design principles ensuring cross-browser compatibility and an accessible, user-friendly interface.

Student Attendance Management App | *Flutter* | *Dart* | *Mobile Development*

- Built a cross-platform mobile application using Flutter for real-time student attendance tracking with an intuitive UI.
- Implemented features for marking, viewing, and reporting attendance, demonstrating skills in mobile app development and UI/UX design.

CERTIFICATIONS & TRAINING

- Python Bootcamp 2025 — Udemy (Verified Certificate)
- Workshop: Roadmap & Careers in Python — Industry Expert Session
- Online Workshop: Robotics & IoT — Hands-on training in embedded systems and automation

RESEARCH & CO-CURRICULAR ACTIVITIES

- Published & Presented: "IoT-Based IV Bag Monitoring and Alerting System" at ICAIC'25 International Conference
- Active participant in tech seminars, coding workshops, and interdisciplinary engineering events

CORE COMPETENCIES & SOFT SKILLS

Communication • Adaptability • Creativity • Leadership • Problem Solving • Time Management • Analytical Thinking • Team Collaboration

DECLARATION

I hereby declare that all the information provided above is true and accurate to the best of my knowledge.

Place: Sivakasi

(Hema Praba D)